

MESO SECTIONS, GUIDING METAPHORS AND OTHERS IN NOTION NOT INCLUDED HERE

HOOVER MICROFOUNDATIONS!

Successful models might resemble the emergence of communication systems more than static money models or current iGSS implementations (Oliphant 1996, Quinn 2011, Arranz 2011)

The Institutionalization of Money: Advancing Fundamental Institutional Theory Through Inverse Generative Social Science – (Short research paper)

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Abstract. Inverse Generative Social Science (iGSS) identifies macro-level patterns and reverse-engineers plausible micro-level decision rules that could have produced them. This paper deploys iGSS in a relational agent-based framework to model the emergence of money as a social institution (i.e., as "shared rules that structure social interaction"). Agents face abstract division of labor problems that can only be solved if they learn to coordinate on rules for distributing goods. A macro-level target of maximizing widespread distribution and consumption of goods across agents is set; through genetic programming, micro-level agents discover decision rules leading to increasingly frequent transfers. We test this setup in three scenarios: one where agents develop rules in isolation, and two relational scenarios where agents are endowed with primitives for prototypical social behavior (spot exchanges and interpersonal debt). Based on preliminary tests, the relational scenarios give rise to complex rule-based behavior that can be interpreted as the emergence of meso-level institutions—some functionally equivalent to money. This approach generalizes previous models of money, providing insight into microfoundations and the balance between rules and equilibria in institutional formation. By explicitly placing institutions as discovered outcomes in a theory-agnostic framework, this work also showcases the potential of iGSS for advancing fundamental social and economic theory.

Keywords: Inverse Generative Social Science, Monetary Theory, Agent-Based Modeling, Evolutionary Algorithms, Meso-Level Institutions.

1 Introduction and Context

Generative Social Science [1] seeks to understand complex socioeconomic systems by simulating micro-level agent behavior and observing emergent macro-level patterns. In contrast, *inverse* Generative Social Science (iGSS) [2] begins with empirically observed macro-level patterns and, through computational optimization techniques like evolutionary algorithms, searches for plausible agent-level rule sets that could have produced them. Although iGSS has mostly been applied to narrowly defined empirical problems, it holds a broader promise to illuminate foundational questions of the social and economic sciences, including the problem of microfoundations in economics, the role of rules and equilibria in the formation of institutions, and the drivers of different forms of socioeconomic organization.

This paper revisits a well-established tradition of "fundamental models of money" in pure economic theory, examines major critiques raised against these abstract formal models by monetary scholarship in history and sociology, and deploys the theory-agnostic toolset of iGSS to develop a synthetic agent-based framework bridging some of the gaps between these perspectives.

Money is a centerpiece of the socioeconomic world, a key aspect of the micro-macro link in economics, and arguably the core social institution enabling all other forms of economic organization. Among the many competing views of the origin of money, a common thread is the observation that economies where money is widely used tend to support higher levels of division of labor—and thus greater economic complexity—than those relying solely on direct interpersonal relations.

We develop an agent-based model posing division of labor problems that can only be solved if agents learn to coordinate and adopt mutually consistent decision rules for the distribution of goods among them. At its level of abstraction, our setup is compatible with both well-established fundamental models of economic systems and stylized observations from economic history, making no strong assumptions in favor of any one perspective. We then apply an iGSS framework by defining a "high level of division of labor" as a macro-level target and implement evolutionary algorithms at the agent level to discover decision rulesets for agents to distribute goods among themselves. We interpret these shared rulesets as meso-level institutions and argue that some of them can be understood as an emergent version of the institution of money. By treating institutions as outcomes of the modeling process within a theory-agnostic framework, this work provides a new, direct connection between abstract models and fundamental sociological theory, showcasing the potential of iGSS as a general epistemological approach for empirically informed theory development in the social and economic sciences.

This section motivates our approach. The following section describes the model, and the final section presents a preliminary discussion of work-in-progress results and expected future developments. We find that the current implementation of iGSS does not yield sufficiently complex agent behavior—partly because agents do not interact in heterogeneous enough ways to generate rich dynamics, echoing sociological critiques of abstract models of money in economics. To address this, we are endowing agents with two relational primitives: spot exchanges and interpersonal debt (reflecting key lines of

thought on the origin of money, namely commodity and credit theories). We expect that enabling even minimally complex relational behavior will broaden the space of possible rule sets for goods distribution, shedding new light on processes of economic self-organization. Both exchange and debt should serve as scaffolding for the gradual evolution of more complex rules of agent interaction, ultimately leading to an increasingly effective division of labor. Future research may investigate the conditions under which debt and exchange themselves might emerge from basic primitives, beginning with simple transfers or donations of goods among agents.

1.1 Motivation – Theories of the Nature of Money

Economics, sociology, and anthropology remain engaged in an open debate to define, specify, and analyze the institution of money from a theoretical perspective. In many of these discussions, money serves as a paradigmatic case for understanding institutions. While money is not as fundamental as language or as designed as organizations, fundamental monetary theory becomes a testing ground for exploring, testing, and developing more abstract theories of social form. This debate goes beyond the traditional contest between commodity, credit, and state theories of money, addressing issues such as whether institutions are better understood as rules or equilibria [3–6], the roles of emergence, systematicity, and collective intentionality in the origins of money [7–10], and, in mainstream monetary theory, the function of token money as a correlating device and mechanism supporting decentralized economic cooperation [11–15].

Formal economic theory, in particular, has a long tradition of fundamental models. These include projects integrating money into the General Equilibrium framework (e.g., through cash-in-advance [16], overlapping generations [17], or money-in-demand-function [18] setups), other approaches related to the abandoned microfoundations project [19–21], and a series of purpose-built models attempting to explain the rationale behind the use of money based on different abstract premises (including trading post models [22]; double coincidence and indirect exchange [23, 24]; information asymmetries [25]; collective book-keeping [26, 27]; economic decentralization [28]; intertemporal [29] and social [30] frictions; commitment and memory [31]). Since iGSS allows for a more flexible approach less committed to a specific set of premises, the framework we propose below can be considered to generalize many of these contributions.

Outside of economics, the diversity of fundamental theories of money is even greater. Often organized around the idea that “money is a social relation” [32], these critical stances emphasize aspects of monetary institutions such as agent-to-agent relations (as opposed to agent-to-commodity) [33], debt-credit fundamentals of sociality [34], hierarchy and distributive conflict [35]; emergent, symbol-mediated interaction and the social conditions for abstract units of account [36, 37]; mechanisms for the formation of socioeconomic structure [38, 39]; and broader social forms and processes [40, 41]. Interdisciplinary monetary theory, in fact, forms a hinge point for post-Keynesian, conventions-school, and sociological critiques of mainstream economics [42, 43], with many approaches in sociology and anthropology deriving different

conceptions of the social world and, consequently, different understandings of money. We return to these to propose a more synthetic interpretation in the next section.

As such, understanding the nature of money remains one of the most challenging and open-ended questions in social theory. Money is not merely an empirical artifact but a foundational institution that links micro-level interactions to macro-level outcomes. Its study requires grappling with unobservable historical processes—such as self-organization, institutional design, market evolution, and state intervention—that resist straightforward empirical or experimental investigation. In essence, the most meaningful questions about money pertain to the nature of the institution itself, rather than to its manifestations as currencies. Viewed from this perspective, monetary theory emerges as a vital subfield of abstract social forms theory, where – given enough care to extricate theoretical issues from broader questions of political economy – a wealth of well-grounded models and empirical insights offers a fertile sandbox for advancing our understanding of economic organization.

1.2 A Potential Contribution from iGSS: Explicit Representation of Institutions as Mediators of the Micro-Macro Link

Agent-based models of “the emergence of money” form a relevant subset of abstract economic theories of money. Some of the most rigorous and informative are proposed by Marimon [44], starting out from the “forced absence of double coincidence of wants” and leading to one universal commodity in an evolutionary learning setting, Howitt & Clower [45], demonstrating economic specialization through pairwise coordination via a means of exchange, and Gintis [46] or Iacopetta [47], studying emergent money as part of a generalized dynamic self-organization of decentralized markets. Typically, these models set up forced division of labor and predefine agent decisions regarding when and how to exchange commodities. They generally incorporate little interdependence between agents (often only using probability estimates over other actions) and arguably simply illustrate general ideas better handled by cooperation or coordination theories [48]. Moreover, they often subscribe to a “metallist” or commodity-centered view of money that does not align with historical and anthropological evidence on monetary evolution [34, 35], since histories of money commonly highlight the roles of interpersonal debt and the state—rather than “truck and barter”—as crucial. In other words, these models are not “free from domain prior” [49]. Additionally, lacking explicit engagement with broader sociological and institutional reflections, these models do not clearly specify what constitutes an institution.

The framework we propose aims to address these shortcomings. We begin with a broad definition of institutions as “rules that structure social interaction” [3, 50]: not merely norms or conventions, but relational and habitual social processes that both result from and shape agent decisions. This idea finds a formal equivalent in the notion that “a rule plus its populations constitutes a meso unit” [51–53]—a domain distinct from both macro-level aggregate patterns and the level of individual agents. This meso-level domain should be understood not in terms of typical social formations like networks or groups, but as the space of shared decision rules adopted by multiple agents simultaneously. Given a set of agent decision rules related to other agents and the

environment, specific forms of organization are likely to emerge. In this view, network and group formations become epiphenomena of rule-structured interactions, both of which require explanation.

Closely related to these concerns, both evolutionary and post-Keynesian economics stress the need for *macrofoundations* in *microeconomics* [21, 54, 55]: economic interpretation should begin with observed macro patterns and then identify the micro-level processes that can plausibly have produced these patterns. In other words, researchers observe only macro patterns (e.g. stylized facts), not the underlying decision rules, but emergent rules must align with pre-identified aggregate patterns and be functional enough to sustain the system. This reversal of explanatory logic sidesteps aggregation issues and may offer greater explanatory power than the neoclassical microfoundations project, which is argued to have faltered due to its heavy reliance on bottom-up ontologies that overlook the constitutive role of institutions [56, 57]. More broadly, this macro-to-micro reversal parallels the epistemology of inverse generative social science, and working with meso-level institutions can provide the key missing concept to mobilize iGSS to advance socioeconomic theory. This makes iGSS an ideal framework for grounding new formal approaches dedicated to studying institutions and social organization. At the very least, it offers a new formal perspective on sets of shared decision rules among agents that could be seen as equivalent to institutions observed in the empirical world.

This paper seeks the simplest possible abstract model framework to operationalize these insights.

2 A proof-of-concept model

The model is a generalized Sugarscape-like setup where agents can produce and consume a variety of different “goods.” Agents are spatially separated, with different agents producing different goods, and all benefit from the consumption of all goods. Agents can recognize and interact with each other, and goods can be transferred, though such transfers are limited by proximity and time constraints. Overall production is sufficient to meet all demands—provided that agents can learn distribution routines that allow one agent’s good to reach another. We define a “high level of division of labor” (i.e., the widespread consumption and distribution of a diverse set of goods by many agents) as a macro-level target, and we implement evolutionary algorithms at the agent level to discover decision rule sets for a widespread distribution of goods. Simulations run until the system converges on a sufficiently high level of division of labor. We then evaluate which emergent rules might correspond to foundational economic institutions such as money..

The table below outlines how this setup can be operationalized in an iGSS framework, following the guidelines provided in Epstein (2023) [2]. Agent primitives are chosen broadly to leave room for multiple views of money from economics and sociology (commodity, debt, units of account). Goods are explicitly represented, have a specific origin in space (a “production site”), and must traverse through agents (beyond mere transaction costs) to be consumed. Both agent-to-agent and agent-to-good

interactions are necessary for the distribution of goods, though we make no strong assumptions about how these interactions must occur.

In contrast to most fundamental models of money in the literature, this is a largely “naturalistic” setup that aligns with some stylized facts of monetary history (see, however, different usages of the term “naturalism” in relation to money [6]). Unlike nearly all previous fundamental models of money:

- We impose no forced absence of double coincidence of wants (where agents are forced to forgo consuming certain products, thereby inducing them to hold certain commodities as a means of exchange).
- Instead of random matching, agents are distributed in space or another structured topology—a more plausible setup for the origin of money from customary relations and other institutions [38]
- Specialized agents can produce an excess of goods, and it benefits all of them to find ways to distribute these surpluses; thus, the focus is on surplus distribution rather than scarcity. This perspective aligns with a “social provisioning” view, where agents find collective means to provide for their needs [58, 59].
- “Production” is situated in time and space: the environment’s structure both enables and constrains access to goods, revealing patterns in which institutions may directly mediate the movement of goods. This leaves room for ideas of money promoting forms of “time-space stretching” in the economy [39], as abstracting away spatial structure might diminish explanatory power.
- We make no strong initial assumptions about agents (e.g., valuation through price vectors, ranking of goods for indirect exchange, or rationality, as in other models). The staged inclusion of specific relational modules (indirect trade, credit-debt rules) allows us to test the effects of adding only certain types of relations, compatible with various theories of the origin of money.
- We make no general assumptions about the general institutional structure of the economy (markets, centralized distribution, and gift-giving are all compatible with historical and sociological notions of money).
- There are no pre-set abstract game payoffs beyond achieving consumption of goods (although any setup is translatable into more abstract game-theoretical terms, such as in a model-of-the-model approach).
- Agents are treated as flexible rule-followers (and rule-learners); unlike extant models, we consider explicit relations both between agents and their environment or goods, and among agents themselves. In a novel sense, these are “institutional” agents (other-regarding, rule-following, and processual).

We use as a macro target the increase in the total quantity of goods consumed by all agents at the end of every period, which serves as a proxy for general welfare. Given the setup, this increase results directly from an overall, integrated division of labor achieved through the distribution of goods. Alternative, less general approaches might define “monetary trade” patterns as in previous models and investigate whether agents can learn compatible decision rules. Theoretical or empirical patterns of division of labor could also serve as references, but this might no longer qualify as a candidate explanation for the “nature of money,” since available macro data is relatively modern and presumes monetary trade. Within this setup, iGSS should enable the emergence of

rule-mediated division of labor, and we seek to identify rule populations that can be interpreted as the use of money.

Table 1. Outline of iGSS Model: Primitives, Operators, Procedures.

Component	Description
Model Target	High levels of product consumption by agents (proof-of-concept; ideally, theoretical or empirical monetary patterns Location and time* Bounded-length inventory of goods (replenished automatically based on location) Bounded-length list of other agent IDs Auxiliary type tags
Agent Primitives	At least one global macro-state variable Interpersonal operations: “give” (inventory transfer); “spot exchange” (conditional inventory swap); “interpersonal debt” (IOUs from one agent to another) Topology: agents fixed in space, interaction within range (translatable into random graph with differentiated agents, at loss of explanatory power) if-then = < >
Combinators	not and or + - * / ”has” (inventory function; necessary for interpersonal interaction) mutate, crossover
Fitness Metric	Number of different items consumed by all agents per time step
Evolutionary Program	Evolutionary Model Discovery [60]
Stopping Rule	Loss value threshold

Finally, we must define what qualifies as “money” in the model. In the wider macrosociological view is, the most rigorous and plausible definitions of money as an institution are differ from the intuitive metallist sense of “objects used for transactions” (among other potentially misleading features of the “folk ontology of money” [61]). In various theoretical approaches, money is primarily a generalized unit of account [35, 36, 62, 63], a means of clearing embedded social debts [34, 35, 38], a convention [64, 65], an instrument of delocalization [39, 66], or a generalized social process that structures the individuals involved with it [36, 38, 67, 68]. Drawing on interdisciplinary monetary theory, we define money, as agnostically as possible, as “any set of decision rules shared across agents, regarding commodities or other agents, and enabling either the deferral of economic transactions in time and/or the structuring of more complex chains of production in space”.

“Growing” a goods distribution/transfer system through rule discovery represents a broader scope for iGSS than the more narrowly empirical uses so far. Rich institutional behavior requires specific types of agent interdependence. We test the framework in

three different scenarios. The first, "methodologically individualist iGSS," deploys the evolutionary discovery inverse-generative framework as applied in most previous studies [2]. We show that this approach is insufficient to generate decision rules that are complex and heterogeneous enough to qualify as an economic institution—partially confirming results from the economics literature. In two additional scenarios, we incorporate different "sociological primitives" into agent specifications. In the "institutional iGSS – exchange" scenario, we predefine an elementary spot exchange mechanism allowing agents to swap goods. In the "institutional iGSS – debt" scenario, agents are endowed with the ability to issue a personal IOU in exchange for receiving a good. Future research should also explore a fourth scenario where decision rule sets are evolved at the level of links between pairs of agents, attempting to grow relational primitives such as debt or exchange.

The model's purpose is open-ended theoretical exploration [69], particularly regarding formal representations of institutions as emergent phenomena that serve as the explanandum of social research—that is, institutions are outcomes rather than premises of the model. This forces us to be explicit about an in-model definition of institutions, which is novel in the monetary literature and may require careful further interpretation regarding the correspondence between model abstractions and the empirical world [70]. Paradoxically, while the focus here is less on parametrization or calibration and more on finding in-model equivalents ("institutional digital twins"), we believe this approach yields more realistic results than more narrowly empirical methods. The key insight here is that institutions themselves, in the real world, are abstract constructs enacted by humans and not directly observable—especially considering cases of institutional formation lost in historical time, such as the emergence of money [34, 35].

3 Preliminary Results and Discussion

Model implementation is at a very early stage. Off-the-box iGSS applications do not generate sufficiently rich rule-based behavior, as all agents are characterized by essentially the same set of rules. This reproduces the issues with representative agents in economics that have long undermined the microfoundations project [21] and echoes sociological critiques of formal models relying on asocial, non-relational agents [9, 57]. We are currently experimenting with relational primitives and the selection of mutually consistent rules at the level of pairs of agents—using spot exchange and interpersonal debt, both highlighted as critical for the emergence of money by different theoretical perspectives. Below, we offer qualitative expectations of future results and anticipated limitations.

This approach raises several challenges for inverse generative social science as a tool for fundamental social theory. To be explanatory, iGSS models must incorporate explicit forms of interdependence that go beyond simply learning mathematical parameters for decision-making. Rich institutional behavior—including social roles such as buyers and sellers, functional differentiation (e.g., some agents produce while others trade), and complex contextual rulesets for valuing different goods or relations—is incompatible with "representative agent" approaches. Such social differences likely

exceed what can be captured by varying parameters alone; agents must differ in the concatenations of rules that characterize them. Asocial agents may partly explain the failure of the microfoundations project, while “somewhat social” agents—those using others’ parameters as input—represent an improvement, and heterogeneous architectures [2] might provide further insight into social roles and specialization. Ultimately, however, unlocking the full potential of iGSS for social theory likely requires exploring conditional interrelations between agents, whether emergent or pre-programmed as primitives. Furthermore, a valid explanation might require functional differentiation between agents, and static macro-target endpoints could detract from the dynamic, process-based approach that would make iGSS valuable for social and institutional process theory [71].

Other limitations of this proof-of-concept model include its very broad abstract domain and a lack of clear correspondence to current empirical data on money—only tentatively matching origin/historical data. At best, successful models developed along these lines would be considered “logically meaningful but historically false” [63]. The model directly represents institutions without parametrizing them, suggesting a trade-off between interpretability and explanatory power. Moreover, the approach leaves out other critical elements in theories of institutions or money, such as the roles of coercion and hierarchy, radical uncertainty or novelty, social embeddedness, and more economically complex production.

Despite these limitations, potential novel contributions from this approach include:

- An “institutional iGSS” framework that speaks to three current open debates in fundamental monetary theory: rules-and-equilibrium [3, 4], social ontology [7, 9], and token-mediated cooperation [12, 15].
- A focus on “institutional agents,” “institutional emergence,” and “explicit institutions” that asserts a domain for more explicit formal institutionalism, treating institutions as outcomes of the model.
- Positive results that could offer a new candidate explanation for the emergence of money, aligning more closely with social theory (and, very cautiously, with historical observations).
- Generative explanation applied to a well-studied theoretical issue for which existing empirical evidence is insufficient; here, varying primitives and macro criteria serve as explanans, with emergent rule sets as explanandum.
- To some extent, a “free from domain prior” approach [49] that offers an informative, though not conclusive, tool for assessing competing theories of money.

Finally, beyond abstract formal models of money, other candidates for iGSS-based explanations in monetary theory may offer more empirical approaches with potentially compatible results. These include current experimental data on tokens and cooperation [12, 15], macroeconomic agent-based models that start from stylized macroeconomic facts [72], complementary currencies forming a well-mapped subspace of transactions in a self-contained form of money [73], or historical studies of monetary forms [74].

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